

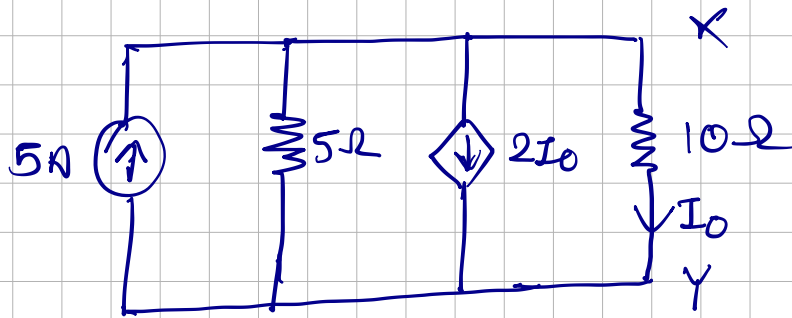
WEEK-3 GRADED ASSIGNMENT

NAME:- SHAHU ROHIT KAMLESH

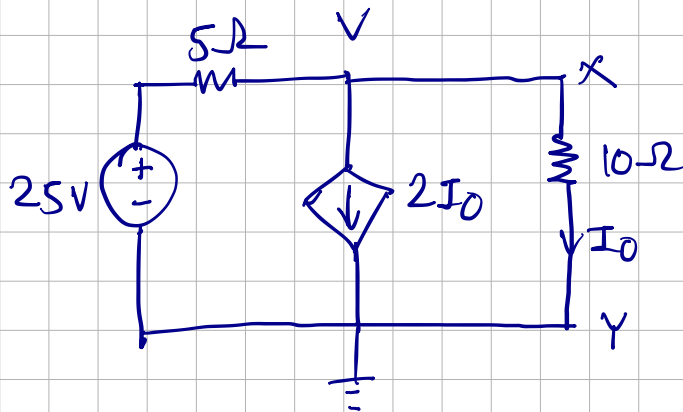
Roll NO:- 25F3100145

Q.1]

Soln:-



Using source transformation.



$$\frac{25 - V}{5} = 3I_0 \quad \text{--- (1)}$$

$$\frac{V}{10} = I_0$$

$$\frac{25 - V}{5} = 3 \frac{V}{10}$$

$$5 = \frac{3V}{10} + \frac{2V}{10}$$

$$5 = \frac{5V}{10}$$

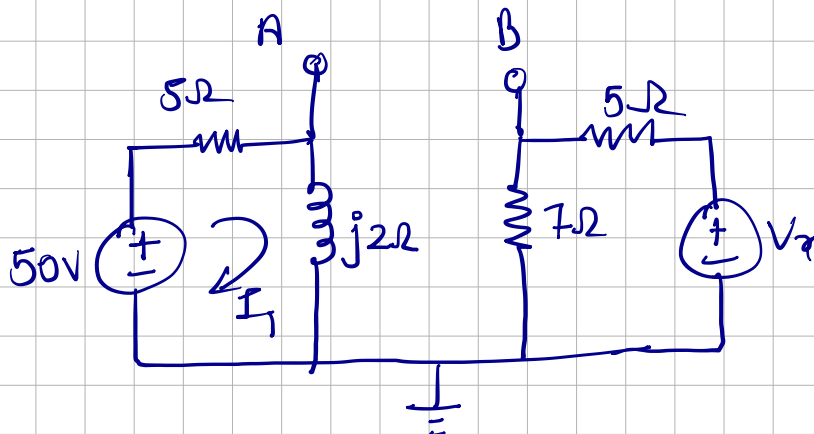
$$V = 10V$$

$$\therefore I_0 = \frac{V}{10}$$

$$I_0 = 1A$$

Q.2]

Soln:-



For Zero current $V_A - V_B = 0$

now
$$I_1 = \frac{50}{5 + j2}$$

$$V_A = \frac{50}{5 + j2} \times j2$$

$$V_A = \frac{100j \times (5 - j2)}{29}$$

$$V_A = \frac{500j + 200}{29}$$

$$V_A = \frac{200}{29} + j \frac{500}{29}$$

now $V_B = \frac{V_x}{12} \times 7$

$$V_A = V_B$$

$$\frac{200}{29} + j \frac{500}{29} = \frac{V_x}{12} \times 7$$

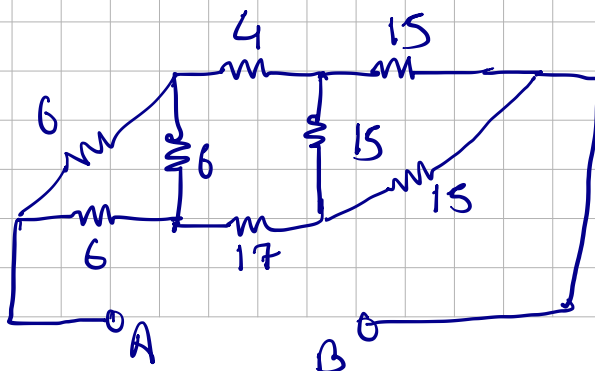
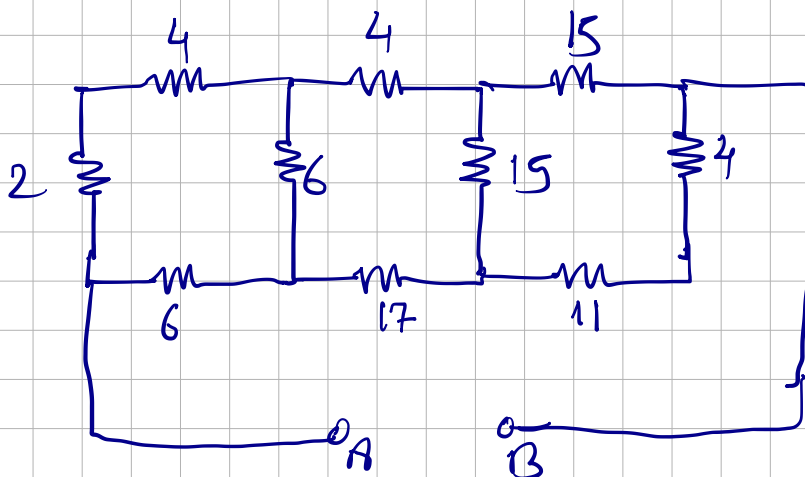
$$\frac{200 \times 12}{29 \times 7} + j \frac{500 \times 12}{29 \times 7} = V_x$$

$$|V_x| = \sqrt{\left(\frac{200 \times 12}{29 \times 7}\right)^2 + \left(\frac{500 \times 12}{29 \times 7}\right)^2}$$

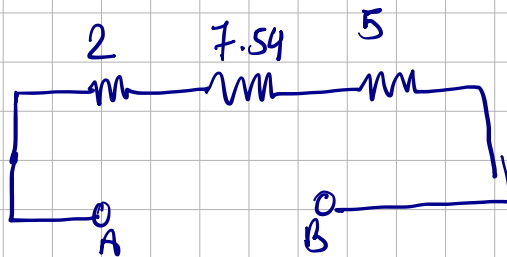
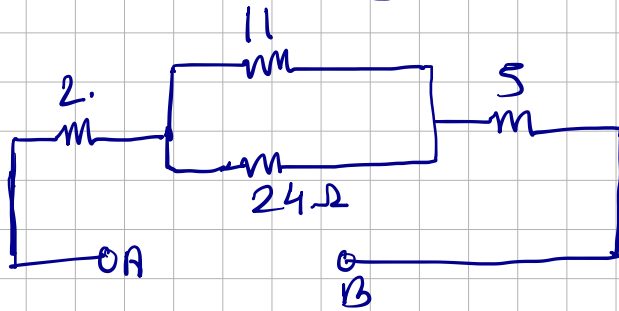
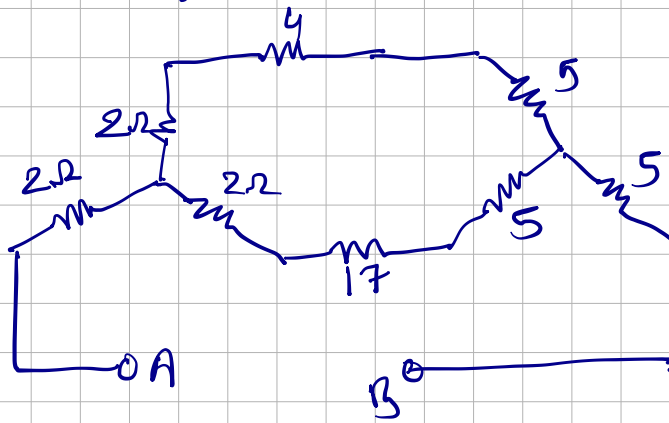
$$|V_x| = 31.833 \text{ V}$$

Q.3]

Soln:-



using star-delta transformation



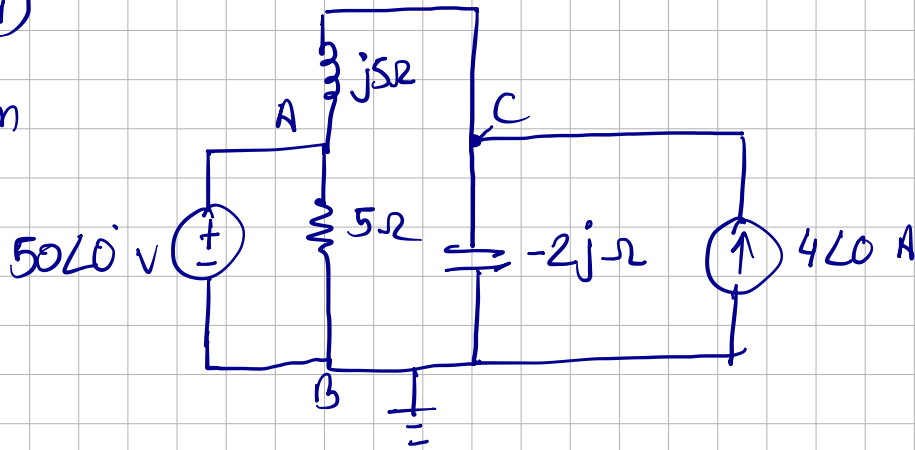
$$\frac{11 \times 24}{35}$$

$$\therefore R_{AB} = 7 + 7.54$$

$$= 14.54 \Omega$$

Q.4)

soln



⇒ Since node A is directly connected to voltage source. & taking B as reference

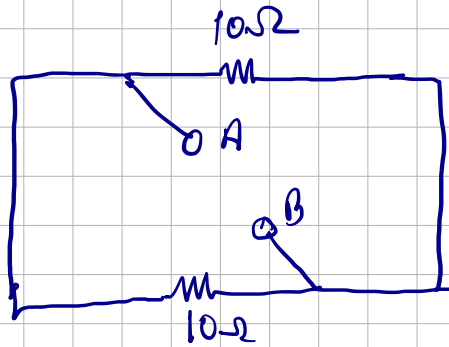
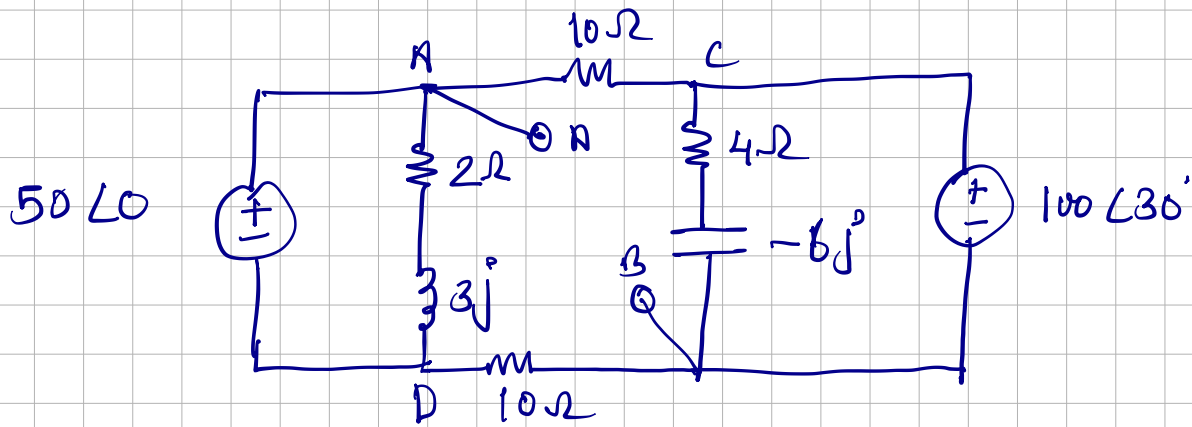
$$\therefore V_{AB} = 50 \angle 0^\circ \text{ V}$$

Q.5]

soln :- phase angle = 0°

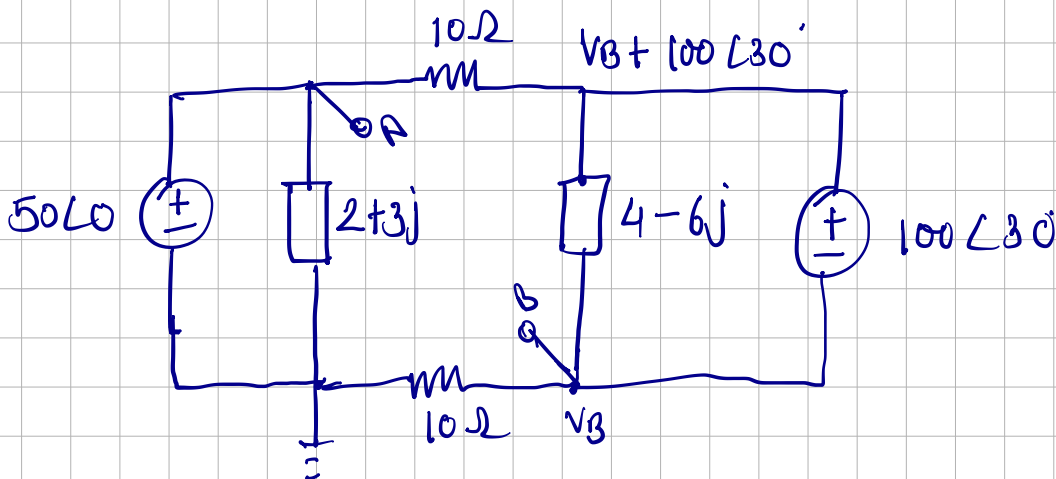
Q.6)

Soln: using Thevenin's theorem.



$$\therefore R_{TH} = 5\Omega$$

Current flowing from AC = Current through B to D



$$\frac{50 \angle 0 - (V_B + 100 \angle 30^\circ)}{10} = \frac{V_B}{10}$$

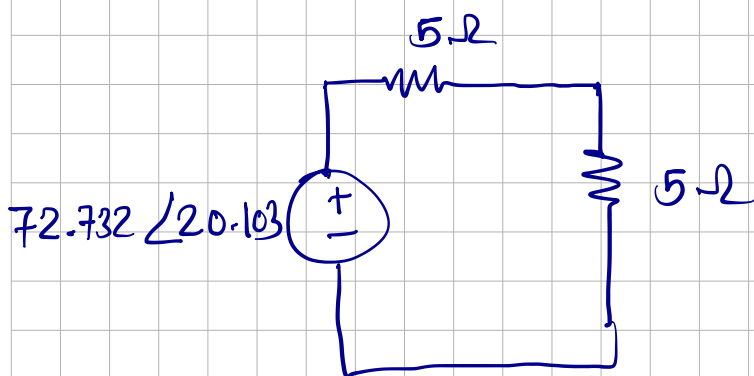
$$50 \angle 0 - 100 \angle 30^\circ = 2V_B$$

$$25 \angle 0 - 50 \angle 30 = V_B$$

$$\begin{aligned} \therefore V_A - V_B &= 50 \angle 0 - 25 \angle 0 + 50 \angle 30 \\ &= 25 \angle 0 + 50 \angle 30 \end{aligned}$$

$$V_{AB} = 72.732 \angle 20.103$$

$$\therefore V_{TH} = V_{AB}$$



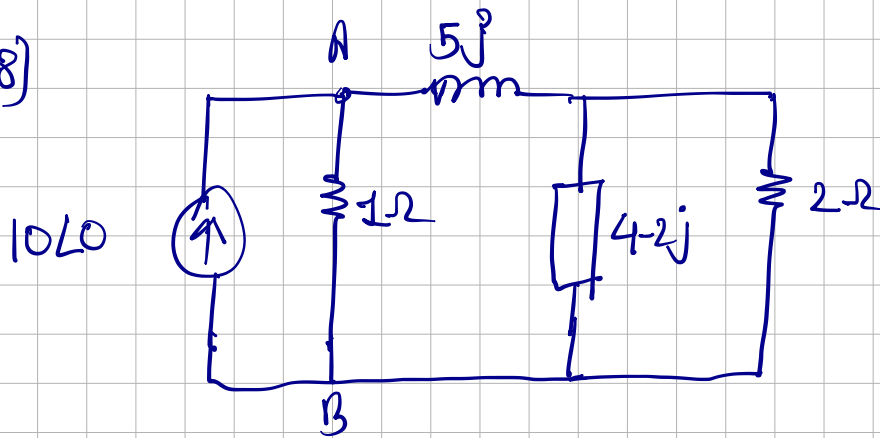
$$\therefore I = \frac{72.732 \angle 20.103}{10}$$

$$I = 7.2732 \angle 20.103$$

Q.7)

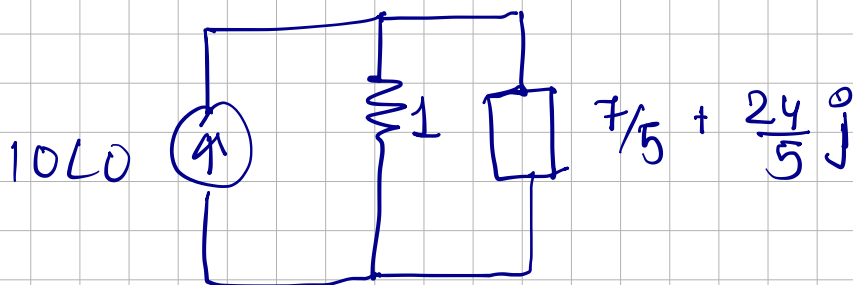
Soln:- $\text{anglu} = 20.103^\circ$

Q.8)



$$\therefore \frac{2 \times (4-2j)}{6-2j} + 5j = Z_{AB}$$

$$Z_{AB} = \frac{7}{5} + \frac{24}{5}j$$



$\therefore$ using current division rule.

$$\frac{10 \times 1}{1 + \frac{7}{5} + \frac{24}{5}j} = I_{j5\Omega}$$

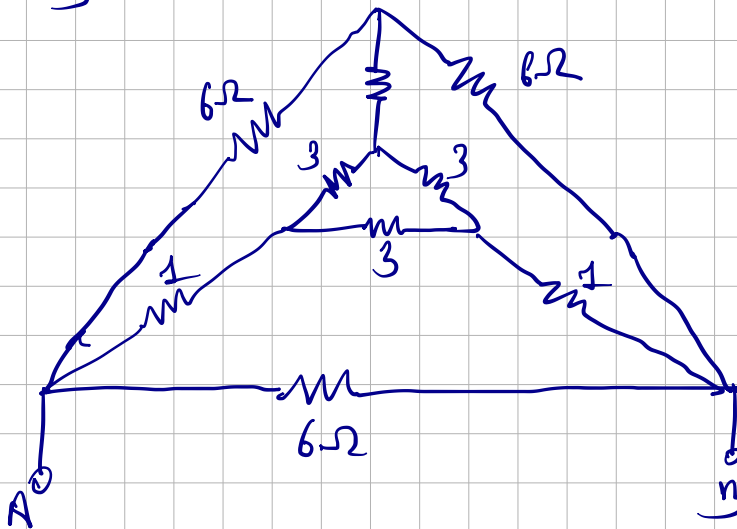
$$I_{j5\Omega} = \frac{5}{6} - \frac{5}{3}j = \frac{5\sqrt{5}}{6} \angle -63.43^\circ$$

$$I_{j5\Omega} = 1.863 \angle -63.43^\circ$$

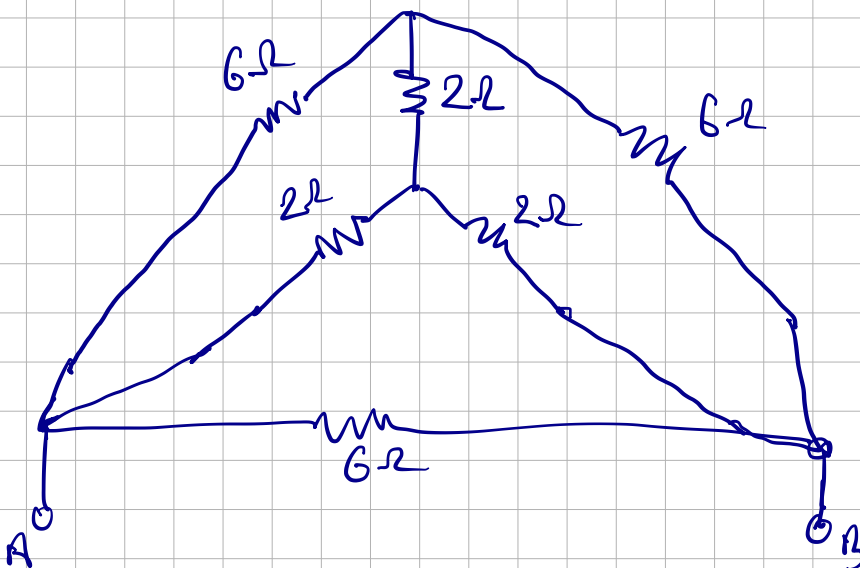
Q.9)

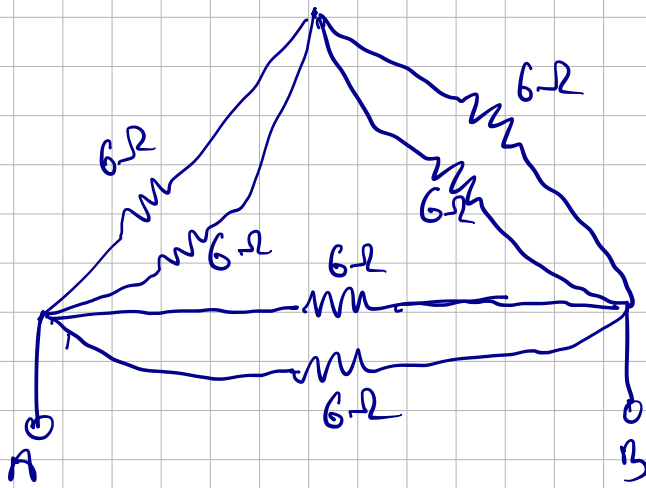
Soln:- Angle = -63.43°

Q.10]

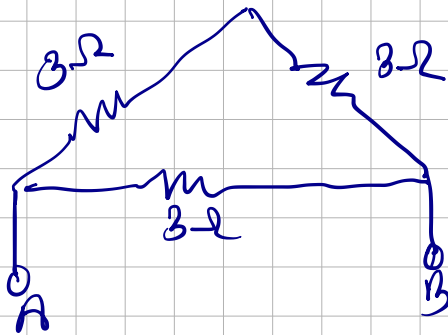


using Star-delta transformation.

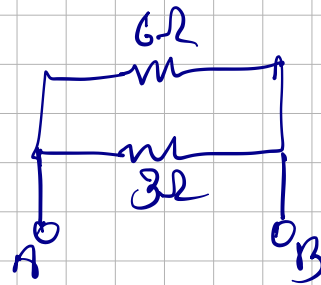




$$2 + 2 + \frac{2 \times 2}{2} = 6\Omega$$



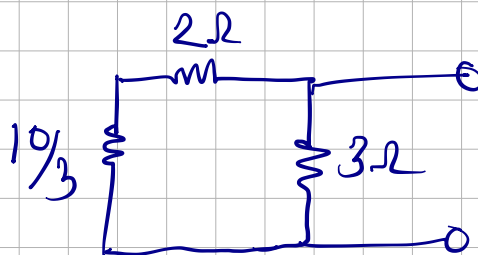
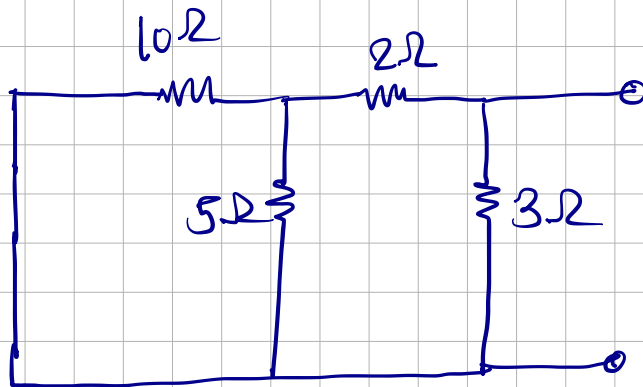
=



$$\therefore R_{AB} = \frac{6 \times 3}{9} = 2\Omega$$

Q.11)

Soln:-



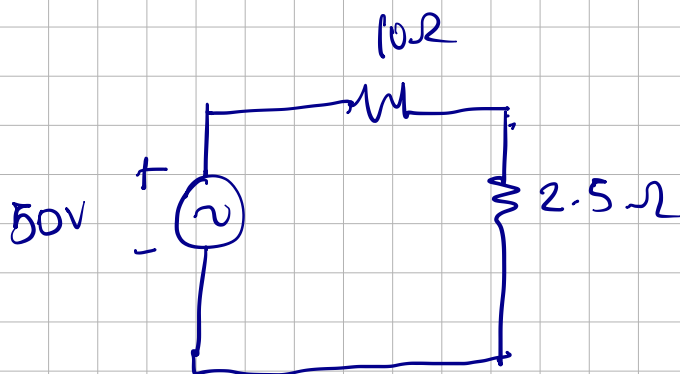
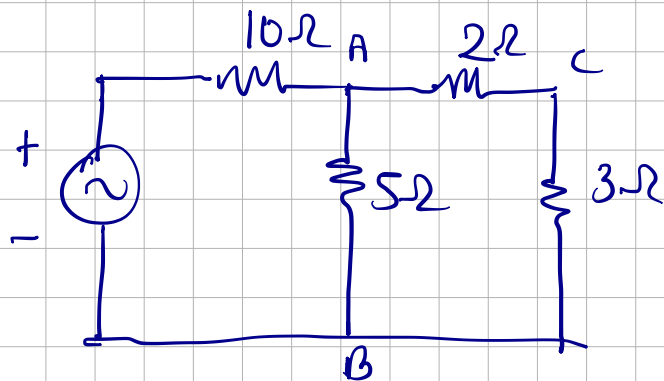
$$R_{TH} = \frac{\left(2 + \frac{10}{3}\right) \times 3}{5 + \frac{10}{3}}$$

$$R_{TH} = \frac{(6 + 10)3}{(15 + 10)}$$

$$= \frac{16 \times 3}{25}$$

$$R_{TH} = \frac{48}{25} \Omega$$

now V_{TH}

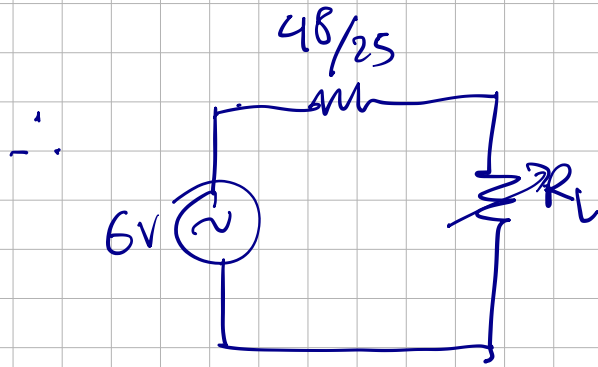


$$\therefore I = \frac{50}{12.5} = 4A$$

$$\therefore \text{Current through } AC = \frac{4 \times 5}{10} = 2A$$

$$\therefore \text{Voltage across } 3\Omega = 2 \times 3 \\ = 6V$$

$$\therefore V_{TH} = 6V$$



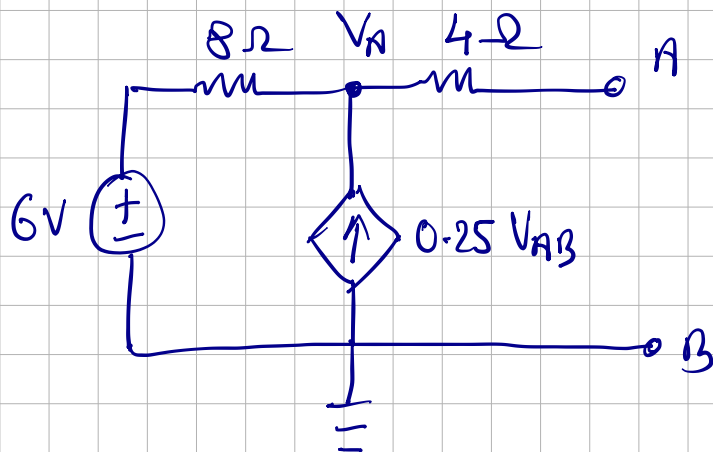
$$\therefore \text{Maximum power} = \frac{36}{4 \cdot \frac{48}{25}}$$

$$= \frac{36 \times 25}{4 \times 48}$$

$$= 4.687 \text{ W}$$

Q.12)

Soln



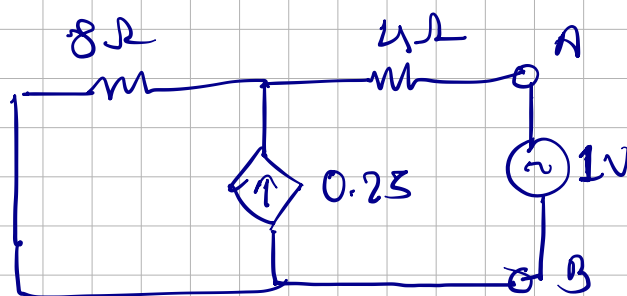
$$\frac{V_A - 6}{8} = 0.25 V_{AB}$$

$$V_A - 6 = 2 V_A$$

$$V_A = -6$$

$$\therefore V_{AB} = -6 \text{ V}$$

Q.13) For R_{TH}



$$\therefore I_{4\Omega} = \frac{0.25 \times 8\Omega}{12\Omega}$$

$$= \frac{0.25 \times 2}{3} = \frac{0.5}{3}$$

$$I \quad R_{TH} = V$$

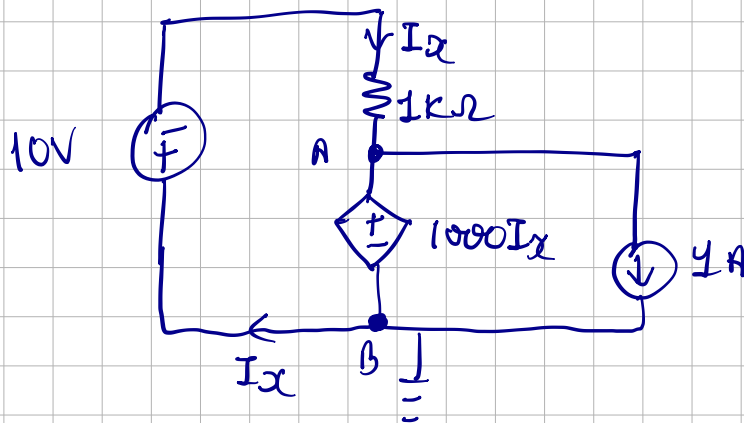
$$R_{TH} = \frac{1}{\frac{0.5}{3}}$$

$$R_{TH} = \frac{30}{5}$$

$$= 6 \Omega$$

Q.14)

Soln:-



$$\therefore 1000 I_x = 1000$$

$$I_x = 1$$

$$\therefore \text{Voltage } AB = 1000$$

$$\therefore R_{TH} = \frac{1000}{1}$$

$$R_{TH} = 1000 \Omega$$